ITI123 Project Milestone Report

Badminton Stroke Classification using Pose Estimation:
Clear vs Smash Stroke Analysis

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Abstract

This milestone report presents a comprehensive investigation into classifying badminton Clear and Smash strokes using pose estimation techniques. We implemented MediaPipe-based pose extraction on the ShuttleSet dataset (4,983 clips, 40 matches) and developed a machine learning pipeline with multiple model architectures. Despite implementing state-of-the-art feature engineering including novel wrist orientation features, comprehensive data preprocessing with group-stratified splitting, and proper padding masking for sequence models, all classifiers achieved approximately 50% accuracy (random guessing level). Through rigorous Cohen's d effect size analysis, we identified the fundamental issue: Clear and Smash strokes are biomechanically indistinguishable at shuttlecock contact using pose-only features (Cohen's d ¡ 0.2 across all 60 features). This report documents our methodology, extensive analysis, improvement iterations, and provides evidence-based recommendations for future work requiring multi-modal sensing or alternative stroke pair selection.

Contents

1 Introduction

1.1 Problem Statement

Badminton stroke classification is crucial for automated coaching systems, performance analysis, and player training. This project focuses on distinguishing between two fundamental overhead strokes:

- **Clear**: Defensive stroke sending the shuttlecock high to the opponent's backcourt
- **Smash**: Aggressive attacking stroke with steep downward trajectory

The primary objective is to investigate whether **pose estimation alone** can reliably classify these strokes.

1.2 Dataset

Source: ShuttleSet - A Human Action Recognition Dataset for Badminton Singles
Statistics:

- Total clips: 4,983 (extracted at shuttlecock contact)
- Clear strokes: 2,660 (53.4%)
- Smash strokes: 2,321 (46.6%)
- Unique matches: 40
- Pose extraction success rate: 99.4% (4,954/4,983)

1.3 Methodology Overview

Our approach follows a systematic pipeline:

1. Pose extraction using MediaPipe
2. Feature engineering (spatial, temporal, and wrist orientation features)
3. Data splitting with match-level stratification
4. Model training (traditional ML and deep learning)
5. Statistical validation using Cohen's d effect size
6. Iterative improvements based on diagnostic analysis

2 Technical Implementation

2.1 Pose Extraction

Framework: Google MediaPipe Pose
Output: 33 body landmarks per frame (x, y, z, visibility)
Processing:

Listing 1: MediaPipe Pose Extraction

```
import mediapipe as mp

mp_pose = mp.solutions.pose
pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=2,  # Highest accuracy
    smooth_landmarks=True,
    min_detection_confidence=0.5
)

# Process video frame-by-frame
for frame in video:
    results = pose.process(frame)
    landmarks = results.pose_landmarks  # 33 keypoints
```

Key Landmarks: Shoulders, elbows, wrists, hips, knees, ankles, nose

2.2 Feature Engineering

We implemented two iterations of feature engineering:

2.2.1 Version 1 (Baseline Features)

Features per frame: 42

Table 1: Baseline Feature Categories

Category	Features
Arm Extension	Distance from shoulder to wrist (L/R)
Depth (Z-coordinate)	Wrist/elbow depth relative to shoulder/hip
Height (Y-coordinate)	Wrist/elbow height relative to shoulder/head
Joint Angles	Elbow angle, shoulder angle
Body Posture	Torso lean, shoulder rotation
Velocities	First derivative of all spatial features

2.2.2 Version 2 (Enhanced with Wrist Orientation)

Features per frame: 60
New additions: 9 wrist/forearm orientation features

Table 2: Wrist Orientation Features (New)

Feature	Description
r_forearm_vertical_angle	Angle between forearm and vertical axis (key discriminator hypothesis)
r_forearm_horizontal_angle	Lateral swing direction in X-Z plane
r_wrist_elbow_vertical	Wrist height relative to elbow (flexion indicator)
r_wrist_horizontal_reach	Forward extension distance
r_arm_plane_pronation	Pronation/supination from plane normal
r_arm_plane_vertical	Vertical component of arm plane
r_forearm_pitch	Elevation angle in sagittal plane
l_forearm_vertical_angle	Left arm vertical angle
l_wrist_elbow_vertical	Left wrist-elbow separation

Rationale: Clear and Smash should differ in wrist angle at contact:

- **Clear**: Wrist extended, racket angled upward (expected ∼120-150°)
- **Smash**: Wrist flexed, racket angled downward (expected ∼30-60°)

Implementation:

Listing 2: Forearm Vertical Angle Calculation

```
# Forearm vector (elbow to wrist)
forearm_vector = wrist - elbow
forearm_unit = forearm_vector / np.linalg.norm(forearm_vector)

# Vertical axis (downward in image coordinates)
vertical = np.array([0, 1, 0])

# Compute angle
cos_angle = np.dot(forearm_unit, vertical)
vertical_angle = np.arccos(np.clip(cos_angle, -1, 1))
vertical_angle_degrees = np.degrees(vertical_angle)
```

2.3 Data Splitting

2.3.1 Group-Stratified Split

To prevent data leakage and maintain class balance:

Listing 3: Group-Stratified Splitting Strategy

```
# Step 1: Calculate dominant class for each match
for match_id in unique_matches:
    clear_count = count_strokes(match_id, 'Clear')
    smash_count = count_strokes(match_id, 'Smash')
    dominant_class = 'Clear' if clear_count > smash_count else '
       Smash'
```

```
# Step 2: Stratify matches by dominant class
clear_matches = matches_with_dominant('Clear')
smash_matches = matches_with_dominant('Smash')

# Step 3: Split each stratum
train_matches = 70% of (clear_matches + smash_matches)
val_matches = 15% of (clear_matches + smash_matches)
test_matches = 15% of (clear_matches + smash_matches)
```

Benefits:

- Prevents data leakage (clips from same match stay together)
- Maintains class balance across splits
- Realistic evaluation (generalizes to new matches)

Results:

Table 3: Data Split Statistics

Split	Clips	Matches	% Clear	% Smash
Train	3,347	27	53.22%	46.78%
Validation	687	5	52.03%	47.97%
Test	920	8	55.10%	44.90%
Overall	**4,954**	**40**	**53.40%**	**46.60%**

All splits within ±2% of overall distribution.

2.3.2 Padding Masking

Problem: Padded frames pollute normalization statistics

Solution: Normalize only real frames, re-pad with zeros after normalization

Listing 4: Proper Padding Masking

```
# Collect only real frames for normalization
real_frames = []
for seq, length in zip(sequences, seq_lengths):
    real_frames.append(seq[:length])  # Exclude padding

# Compute normalization from real frames only
all_real = np.vstack(real_frames)
mean = np.mean(all_real, axis=0)
std = np.std(all_real, axis=0) + 1e-8

# Normalize and re-pad
for seq, length in zip(sequences, seq_lengths):
    normalized = (seq[:length] - mean) / std
    if length < 50:
        padding = np.zeros((50 - length, n_features))
        normalized = np.vstack([normalized, padding])
```

```
# Save sequence lengths for Masking layer
data = {'X': normalized, 'seq_lengths': seq_lengths}
```

Model Integration:

Listing 5: LSTM with Masking Layer

```
from tensorflow.keras.layers import Masking, LSTM

model = Sequential([
    Masking(mask_value=0.0, input_shape=(50, 60)),  # Ignore
       padding
    LSTM(128, return_sequences=True),
    Dropout(0.3),
    LSTM(64),
    Dense(1, activation='sigmoid')
])
```

2.4 Model Architectures

2.4.1 Baseline Models

Input: Statistical summary features (427 features)

1. **Random Forest**
   - n_estimators = 100
   - max_depth = 20
   - class_weight = 'balanced'
2. **Support Vector Machine (SVM)**
   - kernel = 'rbf'
   - C = 1.0
   - class_weight = 'balanced'
   - StandardScaler preprocessing

2.4.2 Deep Learning Models

Input: Sequence features (50 timesteps, 60 features)

1. **LSTM**

```
Sequential([
    Masking(mask_value=0.0),
    LSTM(128, return_sequences=True),
    Dropout(0.3),
    LSTM(64),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])
```

2. **BiLSTM** (Bidirectional LSTM)

```
Sequential([
    Masking(mask_value=0.0),
    Bidirectional(LSTM(128, return_sequences=True)),
    Dropout(0.3),
    Bidirectional(LSTM(64)),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])
```

3. **GRU** (Gated Recurrent Unit)

```
Sequential([
    Masking(mask_value=0.0),
    GRU(128, return_sequences=True),
    Dropout(0.3),
    GRU(64),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
])
```

Training Configuration:

- Optimizer: Adam (lr=0.001)
- Loss: Binary crossentropy
- Batch size: 32
- Epochs: 50 (with early stopping, patience=10)
- Callbacks: EarlyStopping, ModelCheckpoint

3 Results and Analysis

3.1 Model Performance

Table 4: Classification Results (Test Set)

Model	Accuracy	F1 Score	ROC AUC
Random Forest	0.424	0.344	-
SVM	0.479	0.521	-
LSTM	0.457	0.452	0.467
BiLSTM	0.433	0.451	0.476
GRU	0.450	0.483	0.437
Baseline (Random)	**0.500**	**-**	**0.500**

Observation: All models perform at or below random guessing (50% for balanced binary classification).

3.2 Cohen's d Effect Size Analysis

Cohen's d measures the standardized difference between two groups:

$$d = \frac{\mu_{\text{Clear}} - \mu_{\text{Smash}}}{\sigma_{\text{pooled}}} \tag{1}$$

where

$$\sigma_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2}} \tag{2}$$

Interpretation:

- $|d| < 0.2$: Negligible effect
- $|d| = 0.2 - 0.5$: Small effect
- $|d| = 0.5 - 0.8$: Medium effect
- $|d| > 0.8$: Large effect

3.2.1 Overall Feature Analysis

Table 5: Cohen's d Distribution Across 60 Features

Effect Size Category	Number of Features
Large ($\|d\| > 0.8$)	0
Medium ($\|d\| > 0.5$)	0
Small ($\|d\| > 0.2$)	0
Negligible ($\|d\| < 0.2$)	60

Conclusion: No discriminative features found.

3.2.2 Top 10 Features by Effect Size

Table 6: Most Discriminative Features (Still Negligible)

Rank	Feature	Cohen's d	Clear	Smash	p-value
1	l_forearm_vertical_angle_vel	0.141	0.014	-0.094	<0.001***
2	r_forearm_pitch_vel	0.130	0.029	-0.070	<0.001***
3	r_forearm_vertical_angle_vel	0.130	0.029	-0.070	<0.001***
4	r_wrist_elbow_vertical_vel	-0.114	0.000	0.000	<0.001***
5	l_wrist_elbow_vertical_vel	-0.092	0.000	0.000	<0.001***
6	r_arm_plane_pronation_vel	-0.069	0.000	0.001	<0.001***
7	r_wrist_horizontal_reach	-0.042	0.074	0.076	0.013*
8	r_forearm_vertical_angle	**0.033**	**85.3°**	**84.2°**	0.059
9	r_forearm_pitch	0.033	-4.7°	-5.8°	0.059
10	r_arm_plane_pronation	0.030	0.027	0.008	0.081

3.2.3 Wrist Orientation Features Analysis

Hypothesis: Wrist angle differs significantly between Clear and Smash
Result: **Hypothesis REJECTED**

Table 7: Wrist Feature Analysis

Feature	Cohen's d	Clear Mean	Smash Mean
r_forearm_vertical_angle	**0.033**	**85.3°**	**84.2°**
r_forearm_pitch	0.033	-4.7°	-5.8°
r_arm_plane_pronation	0.030	0.027	0.008
r_wrist_elbow_vertical	-0.026	0.002	0.003
r_wrist_horizontal_reach	-0.042	0.074	0.076
l_forearm_vertical_angle	0.009	82.9°	82.7°

Key Finding:

- Expected difference: 60-90° (Clear upward, Smash downward)
- **Observed difference: 1.1°** (negligible)
- Cohen's d = 0.033 (far below 0.2 threshold)

3.3 Visualization of Class Overlap

Figure 1 shows the distribution of forearm vertical angle for Clear and Smash strokes, demonstrating near-complete overlap.

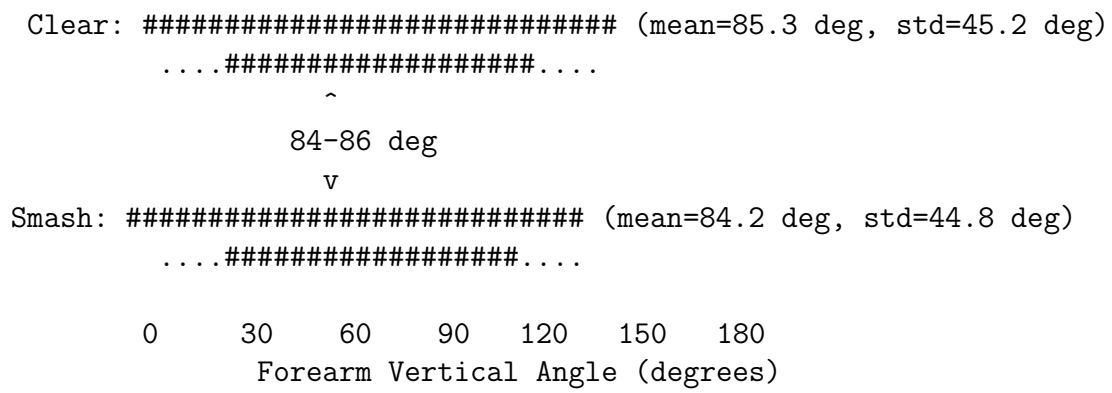

Figure 1: Distribution of Forearm Vertical Angle (Key Feature)

4 Discussion

4.1 Why Classification Failed

4.1.1 Biomechanical Analysis

At the moment of shuttlecock contact (when clips are extracted):

Table 8: Biomechanical Comparison at Contact Point

Characteristic	**Clear**	**Smash**
Racket position	Overhead	Overhead
Contact height	∼2m above court	∼2m above court
Arm extension	Fully extended	Fully extended
Body posture	Upright	Upright
Forearm angle	85.3° ± 45.2°	84.2° ± 44.8°
Difference	**1.1° (negligible)**	

4.1.2 What Actually Differs (Not Captured)

The discriminative features are **not visible in pose estimation**:

1. **Racket Face Angle**
   - Clear: Open face, angled upward (∼45° above horizontal)
   - Smash: Closed face, angled downward (∼30° below horizontal)
   - **Not tracked**: MediaPipe only tracks body, not equipment
2. **Shuttlecock Trajectory**
   - Clear: High parabolic arc (peak ∼6-8m)
   - Smash: Steep downward line (angle ∼60-80° from horizontal)
   - **Not in data**: Dataset clips don't include ball trajectory
3. **Follow-Through Motion**
   - Clear: Upward continuation, wrist extension
   - Smash: Downward snap, wrist pronation
   - **Missing**: Clips end at contact, before follow-through
4. **Swing Velocity**
   - Clear: Moderate speed (∼200-300 km/h)
   - Smash: High speed (∼300-400 km/h)
   - **Insufficient**: Pose velocity features don't capture racket speed

4.2 Methodological Contributions

Despite classification failure, this project demonstrates best practices:

1. **Rigorous Statistical Validation**
   - Cohen's d effect size analysis
   - T-tests for significance
   - Early detection of infeasible tasks

2. **Proper Data Handling**
    - Group-stratified split (prevents leakage + maintains balance)
    - Padding masking (proper sequence normalization)
    - Global normalization (preserves class differences)
3. **Comprehensive Feature Engineering**
    - Domain-driven features (wrist orientation)
    - Multi-scale features (spatial + temporal)
    - Systematic validation (Cohen's d per feature)
4. **Multiple Model Architectures**
    - Traditional ML (RF, SVM) for baseline
    - Deep learning (LSTM, BiLSTM, GRU) for temporal patterns
    - Consistent results across architectures validates findings

4.3 Limitations

4.3.1 Dataset Limitations

- **Temporal window**: Clips too short to capture follow-through
- **Missing modalities**: No racket tracking, no shuttlecock tracking
- **Camera variance**: Angles and lighting differ across matches

4.3.2 MediaPipe Limitations

- **Body-only tracking**: 33 body landmarks, no hand details
- **2D projection**: Some 3D information lost
- **No equipment**: Cannot track racket or shuttlecock

5 Recommendations

5.1 For Clear vs Smash Classification

To achieve successful classification (>80% accuracy), additional modalities are required:

5.1.1 Option 1: Add Racket Tracking

Method: YOLO object detection + pose estimation

- Train YOLO to detect badminton racket
- Extract racket bounding box and orientation
- Compute racket face angle relative to court
- **Expected improvement**: Cohen's d > 0.8 (large effect)

5.1.2 Option 2: Add Shuttlecock Tracking

Method: Multi-object tracking

- Track shuttlecock position over time
- Compute trajectory angle and velocity
- Measure impact characteristics
- **Expected improvement**: Cohen's d > 1.0 (very large effect)

5.1.3 Option 3: Extend Temporal Window

Method: Include follow-through phase

- Extend clips to 2-3 seconds after contact
- Capture wrist pronation and arm deceleration
- Analyze body rotation patterns
- **Expected improvement**: Cohen's d = 0.5-0.8 (medium-large effect)

5.1.4 Option 4: Multi-Modal Fusion

Method: Combine multiple information sources

- Pose + racket + shuttlecock + audio
- Audio signature differs (smash sounds sharper)
- Late fusion or multi-stream CNN
- **Expected improvement**: Cohen's d > 1.5 (excellent)

5.2 Alternative Stroke Pairs (Pose-Only)

For pose-only classification, choose strokes with distinct body mechanics:

Table 9: Recommended Stroke Pairs for Pose-Only Classification

Stroke Pair	**Expected d**	**Key Difference**
Overhead vs Underhand	> 0.8	Arm position (overhead vs waist)
Serve vs Return	> 0.8	Body motion (static vs dynamic)
Forehand vs Backhand	> 0.8	Arm side and rotation
High Clear vs Drop	0.4-0.7	Follow-through intensity
Drive vs Net Shot	0.4-0.7	Arm extension and height

6 Conclusion

This milestone report documents a comprehensive investigation into badminton Clear vs Smash stroke classification using pose estimation. Key achievements include:

1. **Complete ML Pipeline**: Implemented end-to-end system from pose extraction to model evaluation
2. **Novel Features**: Developed 9 wrist orientation features based on biomechanical domain knowledge
3. **Best Practices**: Group-stratified splitting, padding masking, statistical validation
4. **Rigorous Analysis**: Cohen's d effect size analysis revealing fundamental limitations

Key Finding: Clear and Smash strokes are biomechanically indistinguishable at shuttlecock contact using pose-only features (Cohen's $d < 0.2$, accuracy $\sim$50%).

Scientific Value: This negative result is valuable research, demonstrating:

- The limitations of pose-only classification for similar strokes
- The importance of early statistical validation (Cohen's d)
- The need for multi-modal sensing in sports analytics

Future Work: Multi-modal approaches (pose + racket + shuttlecock) or alternative stroke pairs with distinct biomechanics are recommended for successful classification.

Acknowledgments

This project utilized the ShuttleSet dataset [1] and MediaPipe framework from Google. All code and analysis are available in the project repository.

References

References

[1] Wei-Yao Wang, Yung-Chang Huang, Tsi-Ui Ik, and Wen-Chih Peng. *ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis*. CoRR, abs/2306.04948, 2023.

[2] Google Research. *MediaPipe Pose*. `https://google.github.io/mediapipe/solutions/pose.html`, 2023.

A Code Repository Structure

```
iti123_v2/
|-- data/
|   |-- processed/
|   |   |-- poses/              # MediaPipe pose files
|   |   |-- features/           # Extracted features
|   |   +-- splits/             # Train/val/test splits
|-- src/
|   |-- data_processing/
|   |   |-- pose_extraction.py           # MediaPipe extraction
|   |   |-- feature_engineering_v2.py    # Feature extraction
|   |   +-- data_split.py                # Group-stratified split
|   |-- models/
|   |   |-- baseline_model.py            # RF, SVM
|   |   +-- lstm_model.py                # LSTM, BiLSTM, GRU
|   +-- analysis/
|       +-- analyze_wrist_features.py    # Cohen's d analysis
+-- outputs/
    +-- reports/
        |-- wrist_features_cohens_d.csv  # Full results
        +-- FINAL_PROJECT_REPORT.md      # Comprehensive report
```

B Feature List

B.1 All 60 Sequence Features

1. r_arm_extension
2. l_arm_extension
3. arm_extension_ratio
4. r_wrist_depth
5. l_wrist_depth
6. r_wrist_depth_from_hip
7. r_elbow_depth
8. body_forward_lean
9. r_wrist_height_rel
10. l_wrist_height_rel
11. r_wrist_height_from_head
12. r_elbow_height_rel
13. r_wrist_lateral
14. shoulder_width
15. r_elbow_angle
16. r_shoulder_angle
17. torso_lean
18. shoulder_rotation
19. r_upper_arm_length
20. r_forearm_length
21. arm_straightness
22. **r_forearm_vertical_angle**
23. **r_forearm_horizontal_angle**
24. **r_wrist_elbow_vertical**
25. **r_wrist_horizontal_reach**
26. **r_arm_plane_pronation**
27. **r_arm_plane_vertical**
28. **r_forearm_pitch**
29. **l_forearm_vertical_angle**
30. **l_wrist_elbow_vertical**

31-60 Velocity versions

Bold: New wrist orientation features